PRODUCT REQUIREMENTS DOCUMENT (PRD)

Comprehensive System Specification: Student Performance Tracker & Up-Skilling Portfolio Platform

1. Executive Summary & Vision

In contemporary higher educational institutions, academic tracking and career portfolio management operate within disparate silos. While academic offices monitor continuous evaluations, internal examinations, and cumulative grades, students construct their career trajectories via fragmented networks comprising public code repositories, third-party certification sites, and professional resumes. This structural mismatch prevents educational institutions from extracting actionable intelligence regarding pedagogical performance and leaves employers without verified, context-rich evaluations of prospective candidates.

The **Student Performance Tracker & Up-Skilling Portfolio Platform** is a unified ecosystem engineered to bridge this divide. By consolidating continuous assessments, curriculum taxonomies, skill maps, certified verifications, and cross-departmental analytics, the system provides stakeholders with data-driven workflows:

- **Administrators** execute seamless bulk lifecycle procedures, curriculum alignments, and complex academic weighting updates.
- **Students** dynamically contextualize learning outcomes, associate physical proofs with skill proficiencies, and build tailored industry portfolios.
- **Faculty Mentors** leverage granular continuous evaluation analytics to run preventative, data-driven academic and professional interventions.
- **Corporate Recruiters** discover and source verified talent cohorts via highly specialized filtering parameters.

2. System Architecture & Relational Data Model

The platform relies on a heavily relational schema designed to automatically extrapolate career ready indicators from localized classroom metrics. The central design choice maps academic subjects directly to industry-standard **Domains** and individual **Skills**, ensuring that performance in a specific course directly populates a student's technical index.

2.1 Relational Schema Architecture

Entity Table	Key Attributes	Relational Mappings & Constraints
Users	`UserID (PK)`, `Email (Unique)`, `PasswordHash`, `Role (Admin/Teacher/Student)`, `IsActive`	Base authorization entity for all platform system accounts.
Students	`PRN (PK)`, `UserID (FK)`, `Name`, `Phone`, `Batch`, `Division`, `CurrentSemester`, `CGPA`	1:1 mapping with Users. Belongs to specific Batch and Division partitions.
Subjects / Courses	`CourseCode (PK)`, `CourseName`, `Semester`, `Batch`, `DomainID (FK)`	Maps specific curricular courses directly to a centralized Domain entity.
MarkingSchemes	`SchemeID (PK)`, `CourseCode (FK)`, `ExamComponent`, `MaxActualMarks`, `MaxScaledMarks`	Supports scale-down factors for continuous assessment calculations.
MarksEntry	`MarkID (PK)`, `PRN (FK)`, `CourseCode (FK)`, `ExamComponent`, `ObtainedMarksActual`	Atomic record storing raw marks before automated scalar operations.
SkillsMaster	`SkillID (PK)`, `DomainID (FK)`, `SkillName`, `IsCustom`, `CreatedBy`	Hierarchical taxonomy structuring specific technical competencies under domains.
StudentSkills	`StudentSkillID (PK)`, `PRN (FK)`, `SkillID (FK)`, `ProficiencyScale (1-5)`	Maps a student's self-assessed competence to the unified taxonomy.
Certificates	`CertificateID (PK)`, `PRN (FK)`, `StudentSkillID (FK)`, `Name`, `Provider`, `FilePlaceholder`	Provides physical/digital validation for specific skills listed on the profile.

Mathematical Formula for Automatic Mark Normalization

When processing diverse continuous assessment frameworks, the core execution engine isolates the input raw score and computes its scaled equivalent relative to the university master baseline:

$$M_{scaled} = (M_{obtained} / M_{max\_actual}) \times M_{max\_scaled}$$

Where $M_{obtained}$ is the raw unscaled mark submitted by the admin via bulk spreadsheet ingestion, M_{max_actual} represents the maximum achievable raw evaluation score, and M_{max_scaled} is the target constraint defined by the university curriculum template.

3. Comprehensive Module Functional Requirements

3.1 Administrative Operations (Admin Module)

- **Automated Student Provisioning:** The administrative portal handles bulk CSV/XLSX uploads containing basic profile structures. Upon ingestion, the identity service processes the user profile, extracts the target string values, and constructs an initial fallback credential following a strict regex validation mask:

  $Credential = Lowercase(Firstname) + String(BirthYear)$

  The record is committed to the data store with an initial flag state of `IsActive = False` and `ForcePasswordReset = True`.
- **Curriculum & Domain Clustering Engine:** Admins can declare structural course couplings. For instance, *Data Structures and Algorithms (CS-301)* and *Competitive Programming (CS-402)* are grouped under the **DSA/Backend** domain, while *Machine Learning (CS-601)* is grouped under **AI/ML**. This structural mapping allows the analytical engine to build deep competencies across discrete time-series semesters.
- **Flexible Evaluation Layouts:** The framework supports variable, multi-tier assessment schemas. A sample layout like **TL3** can be defined as follows:
  • Continuous Classroom Assessment 1 (CCA1): Max 10 marks
  • Mid-Term Examination: Max 25 marks
  • Continuous Classroom Assessment 2 (CCA2): Max 10 marks
  • Laboratory Continuous Assessment 1 to 3 (LCA1, LCA2, LCA3): Max 5 marks each
  The ingestion pipeline handles raw performance metrics (e.g., scoring 24 out of 30) and applies scaling operations instantly to avoid manual adjustments by staff.
- **Lifecycle Bulk Promotion:** To execute seamless progression updates, the portal provides a tabular grid engine. Admins can select an entire cohort segment (e.g., *Batch 2024 - Division A*), perform a single promotion step ($Semester = Semester + 1$), and clear specific exceptions for individuals flag-marked with active year-downs or cumulative backlogs.

3.2 Student Profile & Professional Portfolio Builder

- **Onboarding & Identity Initialization:** Upon initial login, users encounter a mandatory redirection middleware requiring password updates. The profile activation form forces the user to provide their complete primary bio data: an unstructured "About Me" block, a structured entry for their top 5 secondary school (10th standard) subjects, and a high-resolution, professional portrait. Completion updates the status flags to unlock full platform features.
- **Domain-Aware Skill Graph Mapping:** Technical skill tracking operates strictly within standard corporate domains: **AI/ML**, **CyberSec**, **DSA/Backend**, and **Product Management**. Skill proficiency is declared using a 5-point Likert Scale (1: Novice, 2: Advanced Beginner, 3: Competent, 4: Proficient, 5: Expert).
- **Unified Evidence Architecture:** The portfolio interface uses a smooth modal popup to keep users within their current context. When declaring a skill, a student can click an action link to open a nested panel. This allows them to quickly upload validation documents (such as external training certificates) or attach active code works without leaving the primary view. The asset service generates image snapshots to display as portfolio cards on the public profile.
- **Comprehensive Experience Trackers:**
  • *Industry Internships & Employment:* Tracks paid/unpaid status, organization name, corporate supervisor contact details, faculty guide references, operational model (remote, hybrid, or on-site), and qualitative reflection

blocks.

• *Co-Curricular Engagements:* Mandatory system tracking for institutional activities, association headers, club projects, and event coordination roles to ensure full administrative visibility.

• *Visibility Controls:* Every element has a privacy checkbox toggle. Checking a box updates the attribute flag state to `IsPublic = True`, which includes the item in public talent exports, while unchecking it keeps the content visible to academic mentors only.

3.3 Faculty Mentorship & Analytics Dashboard

The Faculty Module provides deep visibility through a structured, multi-tier interface: **Polytechnic Institute Level → Academic Department (CSE/ECE) → Year Structure → Section Partition (CSE-A) → Individual Student Profile**. This provides teachers with comprehensive metrics to guide student development and make data-driven decisions.

Analytical Visualization	Data Points & Metric Mapping	Mentorship Action Output
Exam Component Trend Analysis	Time-series progression lines tracking averages from $CCA1 \rightarrow MidTerm \rightarrow CCA2$.	Identifies sudden performance drops within specific weeks to trigger early interventions.
Domain Competency Spider Diagram	Radar chart mapping average normalized performance across all grouped course domains.	Highlights functional strengths (e.g., excels in Web/Cloud but requires core support in AI/ML).
Cohort Distribution Histograms	Plots total student counts against performance bands (<40%, 40-60%, 60-80%, >80%).	Evaluates instructional clarity and flags complex modules requiring review sessions.
Cross-Semester Growth Tracker	Longitudinal display mapping individual GPA curves against class performance trends.	Flags downward academic trends across semesters to prioritize mentoring support.
Skill Gap Heatmap Matrix	Cross-references desired target competencies against self-declared student skills.	Identifies skill gaps across cohorts to help design targeted technical workshops.

3.4 Talent Aggregator & Recruiter Interface

- **Verified Cluster Sharing:** Faculty members can select specific students or entire cohorts using precise filters (e.g., *Third Year students with Advanced Python and ≥ 8.5 CGPA*) and generate an isolated, secure access link for employer networks.
- **Redacted Public Resumes:** The external recruiter portal removes internal academic noise. It automatically hides detailed continuous assessment itemizations (such as CCA1 or LCA3 scores) and presents a clean, professional profile focusing on verified core skills, portfolio code bases, past internships, and overall institutional CGPA milestones.

4. Technical & Non-Functional Requirements

- **Security & Access Controls:** Role-Based Access Control (RBAC) enforces strict data isolation. Students are blocked from viewing peer records, and faculty data access is strictly limited to their assigned academic tracks and departments.
- **Audit Logging & Resiliency:** The database maintains complete history records for all bulk data updates, mark changes, and system configuration modifications. The upload engine processes data updates asynchronously to ensure smooth performance during heavy use.